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What is an Operating System?

- OS is a a program that acts as an **intermediary** between a user of a computer and the computer hardware.
- OS is a **resource allocator that** manages all resources, decides between conflicting requests for efficient and fair resource use
- OS is a **control program that c**ontrols execution of programs to prevent errors and improper use of the computer

An operating system is similar to a government. Like a government, it performs no useful function by itself. It simply provides an environment within which other programs can do useful work.

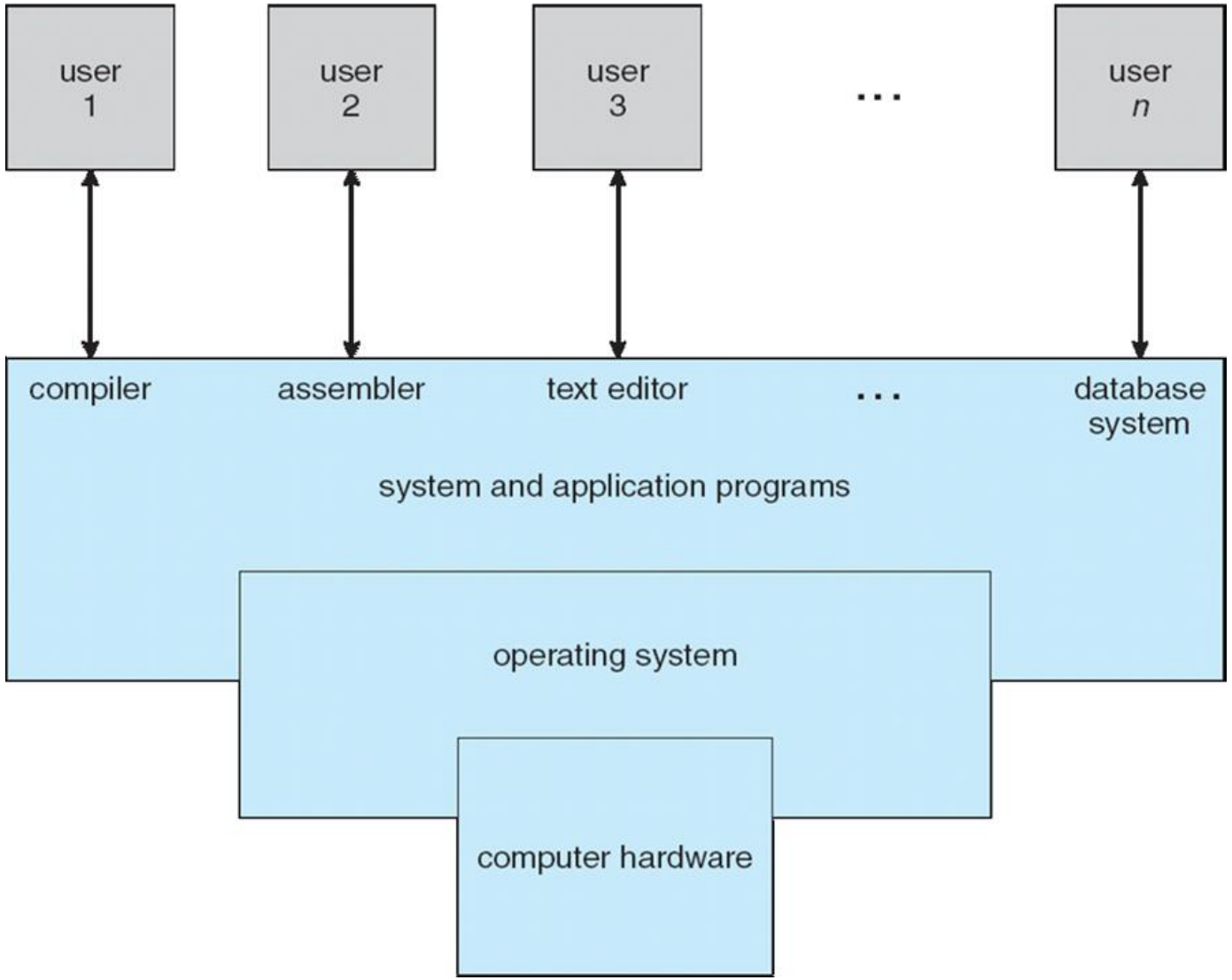
It provides a platform where other application programs can be installed.

Computer System Structure

Computer system can be divided into four components:

- **Hardware –** provides basic computing resources**-** CPU, memory, I/O devices
- **Operating system -** Controls and coordinates use of hardware among various applications and users.
- **Application programs –** define the ways in which the system resources are used to solve the computing problems of the users**-** Word processors, compilers, web browsers, database systems, video games
- **Users-** People, machines, other computers

Four Components of a Computer System



Some important terms

1) Bootstrap Program:- The initial program that runs when a computer is powered up or rebooted.

→It is stored in the ROM.

→It must know how to load the OS and start executing that system.

→It must locate and load into memory the OS Kernel.

2) Interrupt: interrupt is a signal that tells the CPU something important has happened and it needs to stop what it's doing and pay attention.

Hardware may trigger an interrupt at any time by sending a signal to the CPU, usually by the way of the system bus.

3) System Call (Monitor call): Software may trigger an interrupt by executing a special operation called System Call.

- When the CPU is interrupted, it stops what it is doing and immediately transfers execution to a fixed location.
- The fixed location usually contains the starting address where the Service Routine of the interrupt is located.
- Service Routine, where the actions that the interrupt needs to perform are written.
- The Interrupt Service Routine executes.
- On completion, the CPU resumes the interrupted computation

Operating system goals/objectives:

- Execute user programs and make solving user problems easier
- Make the computer system convenient to user
- Use the computer hardware in an efficient manner

Function/characteristics of Operating system –

- **Process Management**: Create, schedule (start/paus/terminate), run, sync, communicate, and end processes.
- **Memory Management**: Track, allocate, free, and protect memory.
- **I/O Device Management**: Control devices, manage data flow, handle errors, and allocate devices.
- **File Management**: Create, organize, access, protect, and back up files.
- **Virtual Memory**: Extend RAM to run large programs using paging and swapping.
- **Security**: Protect data and resources with authentication, authorization, and encryption.